

DATASET

BLS PRODUCER PRICE INDEX FULL TRUCKLOAD TRANSPORTATION

MONTHLY

2003–2018

TRUCKING

LOGISTICS

PPI

SCENARIO

A logistics data science team at a freight brokerage has been tasked with building a **demand forecasting system** for long-distance truckload transportation. Reliable price forecasts are critical: carriers and shippers must negotiate contract rates months in advance, and inaccurate forecasts lead to margin erosion or lost capacity.

The team has access to the U.S. Bureau of Labor Statistics Producer Price Index (PPI) for General Freight Trucking, Long-Distance Truckload (NAICS 484121). This monthly index tracks price changes from the perspective of the seller — the trucking company — and serves as a key leading indicator of freight market conditions.

This is where you come in. You are hired as a data science consultant. Fresh from successfully completing **IM0503 Data Analytics**, you now have the skills and tools to analyse this time series, uncover seasonal patterns and trends, and build forecasting models that can predict future PPI values for the truckload freight market.

DATASET DESCRIPTION

The dataset contains a monthly time series of the BLS Producer Price Index for the long-distance full truckload general freight trucking industry (BLS series ID: PCU484121484121), covering the period **December 2003 to August 2018**. The index is anchored to a base of 100 in December 2003. Note that the dataset provided may contain data quality issues that must be identified and resolved before modelling.

ATTRIBUTE	DESCRIPTION
observation_date	The first calendar day of each month representing the observation period. Acts as the time index for the series. Format: YYYY-MM-DD.
PCU484121484121	The PPI index value for long-distance full truckload general freight trucking (NAICS 484121). Base period December 2003 = 100. Higher values indicate rising producer prices in the trucking sector.

DATA PROVENANCE

Platform: U.S. Bureau of Labor Statistics (BLS) **Series ID:** PCU484121484121 **License:** Public Domain (U.S. Government Data)

Reference:

U.S. Bureau of Labor Statistics. Producer Price Index by Industry: General Freight Trucking, Long-Distance Truckload [PCU484121484121]. Retrieved via FRED, Federal Reserve Bank of St. Louis.

FRED: <https://fred.stlouisfed.org/series/PCU484121484121>

BLS: <https://www.bls.gov/ppi/>